SW Engineering CSC 648-848 Fall 2025

Gator Learn

Team #5

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Milestone 1

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1 Executive Summary

Ever had an experience where you thought "Oh no, I have an exam in a week, and I don't understand the material!" but don't know where to find support? Gator Learn has your back!! Gator Learn is a student-led academic support platform that's designed to empower San Francisco State University students through accessible, affordable and personalized tutoring for SFSU students, by SFSU students. Our platform fosters a collaborative learning environment built on shared academic experiences and community trust. We offer one-on-one and group tutoring across a wide range of SFSU courses including STEM, Humanities, Business, Writing, and general education requirements! Our mission is to enhance student success, confidence and retention by making academic support convenient and relatable. Sessions are available in-person and online with flexible scheduling to fit the demands of university life. Beyond coursework help, tutors provide study strategies, exam preparation, and mentorship to help students thrive long term!

Gator Learn! Helping Gators support Gators!

2 Personae

Persona 1: Sarah Patel - The Student

<u>About Sarah:</u>

- Age: 20
- Major: Computer Science
- Year: Junior

<u>Attitude:</u>

Curious and motivated, but gets easily overwhelmed by heavy coursework and time constraints. Prefers quick and reliable help.

Skills:
Basic to intermediate tech-savvy, uses mobile and web apps regularly,

Limitations:
Struggles with midterms and assignments, and on a right budget.

Goals:
- Find verified tutors from her own university
- Schedule sessions that easily fit her timetable
- Get an affordable tutor to improve the grades..

Pain Points:
- Difficult to find reliable peer tutors quickly
- Limited budget and availability conflict

Persona 2: James Lee - The Tutor

About James:
- Age : 21
- Major: Computer Science
- Year: Senior

Attitude:
Eager to help peers while gaining teaching experience and earn some money.

Skills:
Expertise in specific subjects, and uses digital tools regularly.

Limitations:
Busy schedule balancing coursework, tutoring, and researching

Pain Points:
- Inconsistent tutoring request hours.
- Lengthy verification process for a tutor

Goals:
- Earn extra income through tutoring.

- Build teaching experience for graduate school.
- Connect with motivated students on campus.

Persona 3: Emily Roberts - The Admin

About Emily:

- Role: Academic Success Coordinator
- Age: 38

Attitude:

Responsible for maintaining the quality and smooth operation of the tutoring platform

Skills:

Manages user accounts, approves tutor requests, and oversees system functionality

Limitations:

Managing approval workflows and user issues promptly.

Pain Points:

- Numerous pending tutor applications.
- Limited tools to assess tutor performance efficiently.

Goals:

- Approve and monitor qualified tutors.
- Maintain platform integrity and trust.

Persona 4: Alex Gomez - The Unregistered User

About Alex:

- Age: 18
- Year: Freshmen

Attitude:

Curious, skeptical, and exploring options before committing.

Skills:
Limited to basic web navigation, browsing, and research.

Limitations:
Cannot book the tutors without logging in.

Pain points:
- Unclear service details without registration.
- Concerned about sharing personal university credentials.

Goals:
- Learn what tutoring services are available.
- Understand pricing and reliability before signing up.

3 High-Level Use Cases

Use Case 1: Unregistered User

Alex Gomez hears from his math class about the new SF State tutoring app from his classmates. He will browse around the site, type out his course number into the search bar to see if tutors exist. Alex gets basic information on tutors helping in his courses. He can write a message to the tutor, but is unable to send a message as he hasn't signed up yet. He sees the login and reads the Terms & Privacy, stating university credentials are verified and not sold or shared. **Alex confirms tutors exist for his course and understands he must register to message the tutor with his contact information.**

Use Case 2: Registered User

Sarah Patel hears about the new SF State tutoring app from a flyer on campus. Sarah is desperate and stressed, so she registers as a student in hopes of finding a tutor quickly before her midterm. She registers via student email to confirm she is a student at SFSU. Sarah searches via her course name and course number to find tutors on the subject she struggles with. She finds multiple tutor listings on the subject, but she doesn't know which one is cheapest, reliable, or fits her schedule, so she filters by price from lowest to highest price and is available to meet on

Tuesdays after 2 pm. She finds a tutor who is reasonably priced and can meet at a later time on Tuesdays at the library. She messages the tutor in the app her contact information that she wishes to share, and the rest is history. Tutor and her continue conversation off-site.
Sarah finds at least one tutor matching price/time/location and gets connected.

Use Case 3: Registered User

James Lee hears from other students that they are getting paid to tutor on the SF State tutor app. James does well in the majority of his courses and is looking for a way to make money. He browses the site through courses that don't have as many tutors as he is comfortable with. James first registers as a student with an SFSU email as proof of credentials. He clicks the button to post a listing for every course he thinks he is qualified to tutor for. He attaches his CV to help prove he is reliable to each listing. James has to wait until the admin accepts his listing. He gets accepted, and he gets messages from students about their preferred method of contact. He gets a lot of messages and can choose to ignore them if he is feeling overwhelmed.
James gets his listing approved and sets tutoring rate and times outside the app.

Use Case 4: Admin

Emily Roberts was hired to monitor the app, ensuring students and tutors were behaving. Emily connects to the website database with MySQL Workbench. She gets to check pending posts put up by tutors to ensure nobody posts inappropriate content. She can deny or accept any listing posted. Furthermore, she can check the quality of the tutor via the tutor applying and posting a CV on the listing. Emily receives reports from the public support email on-site in case of inappropriate behaviors from user accounts. She can remove accounts and listings of users who have been reported.
Emily ensures pending posts get processed, and foul acts are restricted from the site.

4 Data Glossary

User Types

1. ***Unregistered User*** - A visitor who can view general information but cannot interact with or book any tutoring services.

   Allowed Actions:

   1.1. View public tutor profiles.

   1.2. Read about the platform and how it works.

1.3. Register using a verified university email to gain full access.

2. ***Registered User*** - A verified university student who has completed registration. They are the main consumers of tutoring services.

   Allowed Actions:

   2.1. Log in with SFSU university email.

   2.2. Search and filter tutors by subject or rating.

   2.3. View tutor's CV, availability, and message on listings.

   2.4. Message tutors once while booking.

   2.6. Post ratings and written reviews after completed sessions.

3. ***Admin*** - Authorized university staff responsible for platform integrity, tutor verification, and issue resolution.

   Allowed Actions:

   4.1. Approve or reject listing verification requests.

   4.2. Manage and remove inappropriate content.

   4.3 Modify system settings or suspend users if needed.

Key Data Entities

Private: These items are visible only to Registered Users after login.

1. **UserAccount**:

   Description: Stores basic information for every individual using the platform , including students, tutors, and admins. .

   Attributes:

   1. Name
   2. E-mail
   3. Password
   4. Live (True/False)

Usage: Used for login, access control, and permission management. Acts as the foundational link between all user-related data (profiles, sessions, messages).

2. **StudentProfile:**

   Description: Holds detailed academic and personal information about registered students.

   Attributes:

   1. Student Name
   2. Major
   3. Year of Study
   4. Session History

   Usage: Used to personalize the student's experience, display relevant tutor recommendations, and maintain a record of their session history and reviews.

3. **Listing:**

   Description: Represents a tutor's service offering

   Attributes:

   1. Student Name (Private)
   2. Major
   3. Subject
   4. Price per hour
   5. Available time
   6. Photo (Optional) (Private)
   7. Resume (CV)
   8. Description
   9. Tutoring Video Sample (Optional) (Private)
   10. Live (True/False)

   Usage: Displayed to students during tutor search and booking. It helps students to compare available tutors based on price, subject, and timing before making a reservation.

4. **Listing Application:**

   Description: Represents the formal application or listing of a student to become a tutor.

   Attributes:

   1. Name

2. Verification Status
3. Reviewed By
4. Decision Date
5. Documents Submitted (Transcript / Resume)

Usage: Used by Admins to review and approve or reject listing applications.

5. **Feedback:**

Description: Captures feedback from students after a tutoring session, including star rating and written comments.

Attribute:

1. Student Name
2. Student Name
3. Rating
4. Comment
5. Posted Date
6. Session Information

Usage: Displayed on tutor profiles to help future students make informed decisions. Used in calculating tutor rankings and performance metrics.

6. **Comments:**

Description: A one-way communication channel where a student sends a message to a listing owner

Attribute:

1. Student Name
2. Tutor Name
3. Message Content
4. Date and Time
5. Phone Number

Usage: Used by the listed student who is getting the tutoring request to get the information about the student, including phone number, and a message..

7. **Subject:**

Description: Represents a course or academic topic that tutors can teach and students can search for.

Attribute:

1. Subject Name
2. Category
3. Description

Usage: Used to categorize tutors and organize search filters by subject area. Supports subject-based matching during tutor discovery.

8. **UserRegistrationRecord:**

Description: Stores registration and verification data for all users.

Attribute:

1. Username
2. Registration Date
3. Account Status

Usage: Used to manage login sessions and enable account recovery.

5 Functional Requirements

Unregistered Users

1. Unregistered users shall be able to register on a dedicated registration page
2. Unregistered users shall be able to register with required name and @sfsu.edu email and optional photo, description, and pronouns
3. Unregistered users shall be able to search public listings by name, price, subject, and course number
4. Unregistered users shall be able to open censored listings

Registered Users

5. Registered users shall inherit all of the above requirements and permissions
6. Registered users shall be able to login on a dedicated login page
7. Registered users shall be able to logout
8. Registered users shall be able to search private listings (instead of public) by name, price, subject, and course number

9. Registered users shall be allowed to edit account information after registering
10. Registered users shall be able to reply to listings with in-house messaging system
11. Registered users shall have access to dashboard that contains messages received and current listings
12. Registered users shall be able to submit listing applications with required title, description, price, and CV and optional photo and video for post approval
13. Registered users shall be able to remove their listings or applications

Administration

14. Administrators shall inherit all of the above requirements and permissions
15. Administrator account registrations shall be assigned by manually website manager
16. Administrators shall review all registrations and listings before they go live
17. Administrators shall be able to view website analytics
18. Administrators shall be able to remove any registered user or listing

6 Non-Functional Requirements

1. Application shall be developed, tested and deployed using tools and servers approved by Class CTO and as agreed in M0
2. Application shall be optimized for standard desktop/laptop browsers e.g. must render correctly on the two latest versions of two major browsers
3. All or selected application functions shall render well on mobile devices (no native app to be developed)
4. Posting of tutor information and messaging to tutors shall be limited only to SFSU students
5. Critical data shall be stored in the database on the team's deployment server.
6. No more than 50 concurrent users shall be accessing the application at any time
7. Privacy of users shall be protected
8. The language used shall be English (no localization needed)
9. Application shall be very easy to use and intuitive
10. Application shall follow established architecture patterns
11. Application code and its repository shall be easy to inspect and maintain
12. Google analytics shall be used
13. No e-mail clients shall be allowed. Interested users (clients) can only message to service providers via in-site messaging. One round of messaging (from client to service provider) is enough for this application. No chat functions shall be developed or integrated

14. Pay functionality (e.g. paying for goods and services) shall not be implemented nor simulated in UI.
15. Site security: basic best practices shall be applied (as covered in the class) for main data items
16. Media formats shall be standard as used in the market today
17. Modern SE processes and tools shall be used as specified in the class, including collaborative and continuous SW development and GenAI tools
18. The application UI (WWW and mobile) shall prominently display the following exact text on all pages "SFSU Software Engineering Project CSC 648-848, Fall 2025. For Demonstration Only" at the top of the WWW page Nav bar. (Important so as to not confuse this with a real application).

7 Competitive Analysis

Feature	[Wyzant]()	[Varsity Tutors]()	[Gator Learn]()
No additional fees	-	-	++
Community Led	-	-	++
Tutor Search by Subject	+	+	+
Tutor Search by Course	-	-	++
Lazy Tutor Applications	+	+	+
Built-in messaging system	++	++	+

++ feature is superior; + feature exists; - feature does not exist

8 System Architecture and Technologies

Server Host: AWS EC2 (t4g.small: 2 vCPUs & 2 GiB of RAM, Free)

Operating System: Amazon Linux 2023

Database: MySQL v8.4 LTS

WebServer: Nginx 1.28.0

SSL Cert: Let's Encrypt (Cert Bot)

SASS: Dart Sass 1.x

Server-Side Language: Java (Amazon Corretto 21)

Web Framework: Spring Boot 3 (Embedded tomcat)

IDE: IntelliJ IDEA

Web Analytics: Google Analytics 4

Build & Run: Maven

Front-End Language: React JS 18.3

Browsers: Chrome and Firefox

9 GenAI Usage

- Used GenAI for creating personae and glossary.

    Prompts & analysis of responses:

1. “Give me a persona image for tutor, admin, students, and unregistered guests for a peer-to-peer tutoring website for a university.” (Highly useful).
   Would not be able to find better than what Gemini gave me.

2. “I am working on a Tutoring website for a peer-to-peer tutoring website for a university. Give me personas containing About, Attitude, Skills, Limitation, and Pain point for each” (Moderately useful)
   The response from Gemini gave me a great start, and helped me to brainstorm ideas on what could be included. Still there were points I had which were better than ChatGPT for Goals and Pain Points.
3. “I am working on the data glossary. It should include the Main actors/Users and the Key entities/terms used in the project.” (Minimally useful)
   The response from Gemini included many feature related terms (e.g, TutorProfile, Two way messaging between students, listed students, and Notifications) which are not core data entities for this project.

10 Team and Roles

Team Lead	Milo Pesce Ares	mpesceares@sfsu.edu
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Support	Samantha Chombo-Rodriguez	schomborodriguez@sfsu.edu

11 Team Lead Checklist

✅So far all team members are fully engaged and attending team sessions when required

✅Team found a time slot to meet outside of the class

✅Team ready and able to use the chosen back and front end frameworks and those who need to learn are working on learning and practicing

✅Team reviewed class slides on requirements and use cases before drafting Milestone 1

✅Team reviewed non-functional requirements from "How to start…" document and developed Milestone 1 consistently

✅Team lead checked Milestone 1 document for quality, completeness, formatting and compliance with instructions before the submission

✅Team lead ensured that all team members read the final M1 and agree/understand it before submission

✅Team shared and discussed experience with GenAI tools among themselves

✅ Github organized as discussed in class (e.g. master branch, development branch, folder for milestone documents etc.)